

2026-2027 Introduction to Network Analysis

Session 1 Handout

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Today's Session

This session will use the Ho Tung family in early twentieth-century Hong Kong to explain what Gephi is and how to interpret a Gephi network. No prior knowledge of Gephi is required, and no coding will be involved. By the end of the session, you will understand the basic uses of Gephi and be able to conduct centrality and network-level analysis of Gephi networks. We will also briefly go through how to perform OCR and create Gephi datasets using IMMARKUS and AI coding agents.

Before the Class

Before your first session, visit the Gephi website and download the latest version of the app to your device. Gephi is frequently updated, but you can continue to use an older version if you already have one on your computer. You can choose to download the Windows or Mac version. After installing Gephi, please also install the **Gephi plugins**. Go to the Tools menu and install GeoLayout, ExportToEarth, and MapofCountries:

Tools > Plugins > Available Plugins > GeoLayout

Tools > Plugins > Available Plugins > ExportToEarth

Tools > Plugins > Available Plugins > MapofCountries

Throughout the course, I will also use IMMARKUS, Codex, and Claude Code. You can download the desktop apps for Codex and Claude Code and obtain an API key for

some of the transcription services on IMMARKUS, but this is **entirely optional**. You can generate a network visualisation and conduct network analysis in Gephi using only Excel and OpenRefine; the latter will be covered in the second session. I have included some AI content in this year's course in response to questions I received from previous participants over the last two years. Instructions on how to obtain API keys for IMMARKUS' transcription services can be found on their GitHub page: <https://github.com/rsimon/immarkus/wiki/05-Annotating-Images>.

If you are interested, you can read the brief history of the Ho Tung family below, but this is not required. **Please bring a computer mouse with you, as you will not be able to navigate Gephi without one.**

A Brief History of the Ho Tung Family

The Ho Tung family comprises three brothers: Sir Robert Ho Tung (1862-1956, 'Robert'), Ho Fook (1863-1926), and Ho Kom-Tong (1866-1950, 'KT'). Robert and Ho Fook are brothers, and KT is their half-brother. Robert and Ho Fook are Eurasian (their father was Dutch and their mother was Chinese), whereas KT is Chinese; he was born after their mother remarried a Chinese man. The Ho brothers were prominent figures in pre-WWII Hong Kong. Their families were known as the Ho Tung family. In the late nineteenth century, as British companies expanded their businesses in Hong Kong, they needed someone who could speak both Chinese and English to act as their intermediary with the locals; this person was known as a compradore. After graduating from Queen's College, the Ho brothers became compradores for British companies such as Jardine Matheson. They were later appointed by the British colonial government as members of the Legislative Council. They also established marital links with other Eurasian families in Hong Kong, such as the Lo, Zimmern, and Wong Kam-fuk families.¹

¹ If you are interested in the history of Eurasians and their communities in pre-WWII Hong Kong, see John M Carroll's *Edge of Empires: Chinese Elites and British Colonials in Hong Kong* (Harvard University Press, 2005) and Emma Teng's *Eurasian Mixed identities in the United States, China, and Hong Kong, 1842-1943* (University of California Press, 2013).

1. What is a Network?

A network is made up of two parts: **nodes** and **edges**. **Nodes** represent entity variables, while **edges** represent the lines that link them.

We need to assign a **weight** to each edge, but there are *no set rules or standards for how to do this. It is entirely up to you and your research topic.* This is entirely up to you and your research topic. In the case of the Ho Tung family, for example, the relationship between brothers is likely to be closer and more intimate than that between half-brothers. Therefore, the weight assigned to the relationship between brothers would be 1, while that assigned to the relationship between half-brothers would be 0.5.

There are two **types of edge** in network analysis: **directed** and **undirected**. A **directed** edge shows a *one-way relationship* between two entities. For example, in a citation network, paper A is cited by paper B. Or in an ownership network, company A is invested in or owns shares in company B. An **undirected** edge, on the contrary, shows a *two-way relationship* between two entities. In our case, Robert Ho is the brother of Ho Fook, while Ho Fook is also a brother of Robert.

2. Drawing a Family Network

IntroNetworkAnalysis repository on github > 2026-27 Session One > Download and open the 'Ho's family_Activity 1' spreadsheet.

Your task is to draw the family networks of Robert and KT by hand on a piece of paper. In addition to drawing the network, try to identify the central figure(s), connectors, bridges, or gatekeepers in the networks.

The spreadsheets illustrate the relationships between individuals. **Figure 1** shows a screenshot of KT's family spreadsheet. Columns A, B and C in the second row are 'Ho Kom-Tong', 'Ho Sui-ting', and 'Brother' respectively, meaning that 'Ho Kom-Tong is the

brother of Ho Sui-ting'. Similarly, columns A, B, C, D and E in the fourth row contain 'Ho Sai-kit', 'Ho Kom-Tong', 'Son', 'Edith McClaymont', and 'Son', meaning that 'Ho Sai-kit is the son of Ho Kom-Tong' and 'Ho Sai-kit is the son of Edith McClaymont', as Edith is KT's wife, as shown in the third row. Each group has 15 minutes to complete this task.

	A	B	C	D	E	F	G
1	EngName	Related to	Relationship	Related to	Relationship	Related to	Relationship
2	Ho Kom-Tong	Ho Sui-ting	Brother				
3	Edith McClaymont	Ho Kom-Tong	Wife				
4	Ho Sai-kit	Ho Kom-Tong	Son	Edith McClaymont	Son		
5	Winnie Choa	Ho Sai-kit	Wife				
6	Choa Leep Chee	Winnie Choa	Father				
7	Ho Sai-wah	Ho Kom-Tong	Son	Edith McClaymont	Son		
8	May Zimmern	Ho Sai-wah	Wife	Andrew Zimmern	Daughter	Ethel Zimmern	Sister
9	Ho Sai-man	Ho Kom-Tong	Son	Edith McClaymont	Son		
10	Ivy Tang Oi-wah	Ho Sai-man	Wife				

Figure 1: Screenshot of KT's Family Spreadsheet

3. Visualising and Measuring Networks with Gephi

Importing a dataset into Gephi:

1. IntroNetworkAnalysis repository on github > 2026-27 Session One > Download Ho Tung family_Gephi.xlsx. **Do not open the spreadsheets in Excel, if you have opened them, close them before opening Gephi.**
2. Open Gephi.
3. Click on 'New Project' and then go to 'Data Laboratory'.
4. Click on 'Import Spreadsheet', then select the Ho Tung family_Gephi.xlsx and press 'open'.
5. Import the 'Nodes table', then go to 'Import spreadsheet' and import the 'Edges table'.
6. Select 'BigDecimal' for the 'Weight' column.
7. Select 'Undirected' for the 'Chart type' and then select **'Append to existing workplace'**.
8. Go to 'Overview' and select 'ForceAtlas' for the layout.

Measuring the networks

1. Go to 'Statistics'.
2. Run 'Average degree', 'Avg. Weighted Degree', 'Network Diameter', 'Modularity', and 'Eigenvector Centrality'.
3. Go to 'Data Laboratory', and you will find statistics for several types of centrality measures, including degree (centrality), weighted degree, betweenness centrality, and eigenvector centrality.
4. Go back to 'Overview' then go to 'Appearance'.
5. Select 'Node', click the 'Colour' button, then select 'Modularity Class' under 'Partition'.
6. Click on 'Apply'.

Interpreting the networks

1. Degree centrality: number of edges linked to each node (popularity, well connected)
2. Weighted degree centrality: Number of edges connected to each node, taking into account the weight of the edges (popularity, well connected)
3. Betweenness centrality: the number of times a node is on the shortest path between other nodes (bridges, gatekeepers)
4. Eigenvector centrality: similar to degree centrality, but takes into account the centrality of the nodes it's connected to
5. Modularity Class: Community detection

Take 5 minutes to discuss with your group about the meaning or how we can interpret the results above. Think about why some nodes (figures in the networks) will have such a high degree of centrality measures in the networks? Based on the results, what were their roles in the networks of the Ho Tung family? Each group has 3 minutes to share their findings with the class.

4. Composing Gephi Datasets

Use the break to think about how to convert the Excel spreadsheets of Robert and KT's families into the spreadsheets that we have just imported into Gephi?

Here are some hints:

1. What are the differences between the Ho family's spreadsheets and the ones we used for Gephi?
2. What do we need to include in the spreadsheets for Gephi?
3. What kind of functions can we use in Excel to create the spreadsheet for Gephi in a more efficient way?

Preparing a Gephi Dataset in Excel (the most challenging step when using Gephi)

1. IntroNetworkAnalysis repository on github > 2026-27 Session One > Download and open the 'Combined' spreadsheet. This spreadsheet shows the family relationships of Robert, Ho Fook, and KT.
2. Copy all the entities in column A to a new spreadsheet, then remove the duplicates (Data > Remove Duplicates).
3. Do the same for column B.
4. Copy the filtered list of column B entities under that of column A, and then remove the duplicates.
5. Add one row to the left and one to the top of the filtered entities. Assign an ID to each entity. Rename the spreadsheet 'Nodes' (nodes table)
6. Go back to the 'Combined' spreadsheet. Copy all the entities in column C ('Relationship') to a new spreadsheet, then remove the duplicates.
7. Assign a weight to each type of relationship. In our case, the default weight is 1, except for 'half brother', 'adopted son', 'adopted brother', 'half sister', 'adopted sister', and 'adopted daughter'. The weight for these relationships is 0.5.

8. Add a column next to columns A and B, then copy the 'nodes table' to this spreadsheet.
9. Use the 'xlookup' function to match the entities to their IDs. For more information on what xlookup is, see <https://support.microsoft.com/en-gb/excel/functions/xlookup-function>.
10. Do the same for the weight of the relationship type.
11. Open a new spreadsheet. Copy the IDs of the entities from column A to the first column. Rename this column 'Source'.
12. Copy the IDs of the entities from column B to the second column. Rename this column 'Target'.
13. Copy the weight of the relationship type from column C to the third column. Rename this column 'Weight'.
14. Add a column next to 'Weight'. Rename it as 'Type'. Since relationships between family members are two-way, enter 'Undirected' in this column.
15. Rename the spreadsheet to 'Edges' (edges table).
16. Copy the 'Nodes' and 'Edges' to a new Excel file and rename it 'Ho Family Network_Gephi'.

5. Composing Gephi Datasets with AI

OCR Printed Materials with IMMARKUS

IMMARKUS is a semantic image annotation tool devised by Hilde De Weerdts and her team as part of the project 'Regionalising Infrastructures in Chinese History'. It is equipped with a range of AI models to perform the OCR for the text on the images.

1. IntroNetworkAnalysis repository on github > 2026-27 Session One > Download Ho Family Chart.jpg.
2. Create a folder on your device. Name it 'Ho Family Chart' and put the 'Ho Family Chart.jpg' in the folder.
3. Open IMMARKUS, <https://immarkus.xmarkus.org/>.

4. Click on 'Open New Folder'.
5. Import the 'Ho Family Chart' folder into IMMARKUS. Choose 'Allow'.
6. Click on the image.
7. Select 'Auto Transcribe' from the drop-down menu (the icon looks like an orange pen).
8. Click 'Select Service' to open the Auto Transcribe interface. The interface includes a zoomable image on the left and a service settings panel on the right.
9. In the service settings panel, select a transcription service from the drop-down list. Some services require an API key or endpoint setup. Visit the IMMARKUS github page for instructions on how to obtain an API key.
10. Set your Endpoint and API key as required, select the image area, then click 'Run Transcription'.
11. Click on Import to IMMARKUS to import the transcription as a note.

For the detailed procedures of steps 6-10, visit the IMMARKUS github page:

<https://github.com/rsimon/immarkus/wiki/05-Annotating-Images>.

Create Gephi Datasets with AI Agents

1. Download Codex Desktop/ Claude Code Desktop app.
2. IntroNetworkAnalysis repository on github > 2026-27 Session One > Download the 'Ho Tung Family Network AI agent test' spreadsheet.
3. Create a folder on your desktop for the following tasks. You can name the folder whatever you like, for example 'Ho Tung Family Gephi Test'.
4. In Codex or Claude Code, select 'New chat' or 'New', then choose the folder that we have just created on the desktop. This is where you can find the Codex or Claude Code output.
5. Enter 'Prompt 1'. Then attach the JPG image of the Ho Tung family tree, which can be found in the 'Ho Family Chart' folder.
6. Select 'New chat' or 'New'.
7. Enter 'Prompt 2'. Attach the 'Ho Tung Family Network AI agent test' spreadsheet.

8. Enter 'Prompt 3' in the same chat.

For a technical explanation of AI agents, see Tang, K.-. leong . "AI Agents for Sinitic Texts". Version 2.0. Zenodo, June 13, 2026. <https://doi.org/10.5281/zenodo.20673640>.

Prompt 1: OCR Text Formatting (the final paragraph of this prompt is IMMARKUS' transcription of Ho Family Chart.jpg. I used Gemini 2.5 Flash to transcribe the text)
(Codex GPT-6.1 Sol: 5m47s; Claude Code Sonnet 5.5: about 2 minutes)

Formatting the following OCR text into an excel spreadsheet, focusing only on the branches of Ho Tung (2), Ho Fook (3), Ho Kom-Tong (5), and Kwok Sui-ting (7). Put their English names in the first column of the spreadsheet and their Chinese names in the second. When formatting the spreadsheet, ignore the branches in the original chart and enter all the English names in column A and their respective Chinese names in column B.

Many individuals with the surname Ho (何) in the chart are written without their surname. For example, 'Sai-wing', 'George', and 'Sai-kwong', should be entered into the spreadsheet with their surnames, such as 'Ho Sai-wing', 'George Ho', and 'Ho Sai-kwong'. Do the same for their Chinese names, for example write '世榮' as '何世榮' and '佐芝' as '何佐芝'. Many individuals with the surname Ho are written in the format 'English given name, Chinese given name', for example 'Edward, Sai-kim' and 'Grace, Yiu-chee'. Reorder them in the spreadsheet as 'Chinese surname Chinese given name English given name', for example 'Ho Sai-kim Edward' and 'Ho Yiu-chee Grace'.

Finally, correct any OCR errors based on the attached image.

C.H.M. Bosman - Shi Tai 施娣 - Kwok Hing-yin 郭興賢 1 Ho Pak-ngan 何柏顏 m Tsoi Sing-nam 蔡星南 1s 1d Tsoi Po-sang 蔡浦生 Tsoi Kit-ching 蔡潔清 2 Ho Tung 何東 5s 8d Sai-wing 世榮 Henry, Sai-kun 世勤 Edward, Sai-kim 世儉 Robert, Sai-lai 世禮 George 佐芝 Victoria, Kam-chee 錦姿 Daisy, Wai-Chee 慧姿 Mary, Shun-chee 純姿 Eva, Han-chee 嫻姿 Irene, Ki-chee 綺姿 Jean, Man-chee 文姿 Grace, Yiu-chee 堯姿 Florence, Hau-chee 孝姿 3 Ho Fook 何福 7s 5d Sai-wing (adopted by Ho Tung) 世榮 Bessie, Bo-chi 寶姿 Sai-iu 世耀 Sai-kwong 世光 Sai-leung 世亮 Sai-chuen 世全 Sai-cheuk 世焯 Elizabeth, Bo-yung 寶容 Sai-ki 世奇 Victoria, Bo-lin 寶蓮 Nancy, Bo-ji 寶芝 Phyllis, Bo-yin 寶賢

4 Ho Moon 何滿 adopted by Pau 鮑 No Issue 6 Ho Yau-kai 何佑啟 aka Walter Bosman No Issue 5 Ho Kom-tong 何甘棠 15s 14d Elizabeth, Pak-ling 柏齡 Sai-kit 世傑 Elsie, Pak-ching 柏貞 Sai-wah 世華 Sai-man 世文 Sai-cheong 世昌 Sai-on 世安 Sai-lok 世樂 Flora, Pak-kin 柏堅 Pak-yin 柏賢 Sai-hong 世康 Sai-kin 世健 Pansy, Pak-siu 柏韶 Rebecca, Pak-fong 柏芳 Sai-wai 世威 Sai-yung 世勇 Genevive, Pak-tuen 柏端 Sai-hung 世雄 Nancy, Pak-shui 柏瑞 Sai-kong 世剛 Sai-keung 世強 Paul, Sai-mang 世猛 Rose, Pak-sheung 柏嫦 Stella, Pak-ngor 柏娥 Sai-lit 世烈 Pak-bun 柏奔 Monica, Pak-woon 柏媛 Grace, Pak-yung 柏容 Pak-ngan 柏顏 7 Kwok Sui-ting 郭瑞亭 aka Ho Sui-ting 何瑞亭 aka Ho Pak-bun 何柏奔 m Wong Kam-fuk 黃金福 6s 3d Wong Sik-lam 黃錫霖 Wong Sik-to 黃錫滔 Wong Sik-man 黃錫文 Wong Sik-chung 黃錫松 Rose Wong Wai-yin 黃煒賢 Pansy Wong Wai-yu 黃煒茹 Wong Sik-kuen 黃錫權 Jasmine Wong Wai-sheung 黃煒嫦 Wong Sik-hon 黃錫漢

Prompt 2: Create a Gephi Dataset

(Codex: 6m43s; Claude Code Sonnet 5.5: 1m16s)

Create a Gephi dataset using the attached spreadsheets. The 'Index' spreadsheet contains the nodes in the network. Leave this untouched and just rename it 'Nodes'. The 'Combined' spreadsheet shows the relationships between the nodes (the edges). For example, columns A, B, and C in the second row contain 'Sir Robert Ho Tung', 'Ho Fook', and 'Brother', meaning that 'Sir Robert Ho Tung is the brother of Ho Fook'. Similarly, columns A, B, and C in the third row contain 'Sir Robert Ho Tung', 'Ho Sui-ting', and 'Half-brother', meaning that 'Sir Robert Ho Tung is the half-brother of Ho Sui-ting'. As Gephi cannot read characters, rename the people in columns A and B using their IDs from the 'Index' spreadsheet. For example, 'Sir Robert Ho Tung' is '1', 'Ho Fook' is '33', and 'Ho Sui-ting' is '23'. Column C shows the relationships between these individuals, replacing the relationship types with 'weights'. The default weight for all edges is 1, but use 0.5 if the relationship type contains the words 'adopted' or 'half'. Add a new column to the right of column C and enter 'Type' in the first row. Since relationships between family members are two-way, enter 'Undirected' in this column. Finally, replace 'EngName', 'Related to', and 'Relationship' with 'Source', 'Target', and 'Weight', and rename the 'Combined' spreadsheet 'Edges'.

Prompt 3: Interpreting the Network

(Codex: 11m53s; Claude Code Sonnet 5.5: 1m40s)

Using the Gephi dataset, identify the clusters in the networks and rank the nodes according to four centrality measures: degree centrality, weighted degree centrality, betweenness centrality, and eigenvector centrality.

Before the next class:

Think about your own Gephi project and try to collect data for it. Everyone will have some time to talk about your project in the next class.